Asymptotic Analysis & Time Complexity

Asymptotic Analysis

Q)

Explain the time complexity of the following code snippet in regards of the Big-O notation:

```
 for (i=0; i<n; i+=4) {
        for (j=1; j<n; j*=2) {
                for (k=0; k<30; k++) {
                        print("Am I still not 30?!!");
                }
                print("Why, God, why? We had a Deal!");
                for (m=n; m>0; m-=2) {
                        print("Could you BE more dramatic?");
                }
        }
}
```

Q)

In the primary scholarship exam in Bangladesh, four lakh (n=4,00,000) students take part but only the top 50 students are given an award.

Write the asymptotic time complexity to give the awards. Assume that each award is given in a constant time.

Q)

Write the asymptotic time complexity of the following function.

```
def contains_duplicates(elements):
  for outer in range(len(elements)):
     for inner in range(len(elements)):
        if outer == inner:
           continue

        if elements[outer] == elements[inner]:
           return True

return False
```

Q)

Express the following running time T(n) with an asymptotic bound.

$$T(n) = 625T(n/5) + n^3$$

Any method is acceptable as long as you show calculations.

Q)

Write the asymptotic time complexity of the following code snippet. Show your works/reasoning.

```
for i in range (1,n)
    j= 1
    while j < i*i
        j= j+1
```

Q)

Analyze the code snippet and answer the following questions:

```
int func(int n) {
    if (n <= 1) {
        return 1;
    } else {
        int a = 0;
        for (int i = 0; i < 3; i++) {
            a += func(n / 2);
        }
        int b = func2(n);
        return a + b;
    }
}

int func2(int n) {
    int c = 0;
    for (int k = n; k >= 1; k -= 4) {
        for (int m = 0; m < n; m += 2) {
            c += k + m;
        }
    }
    return c;
}
```

a) Write the recurrence relation of the following function "func".
b) Calculate the time complexity of the recurrence relation using any of the methods.

Q)

a) For each of these pairs of functions $(f(n),\ g(n))$, identify whether $f(n) = O(g(n))$ or $f(n) = \Omega(g(n))$.

   Show your calculation logic in the script and Tick only the most specific one of the two options in each case in the question paper.

$f(n)$	$g(n)$	$f(n) = O(g(n))$	$f(n) = \Omega(g(n))$

3^n	$2^n . n^4$		
$(\sqrt{n} + n)(30\sqrt{n})$	n^2		

b) Consider the recurrence relation $T(n) = T(\sqrt{n}) + O(n)$. Can we solve this using the Master Theorem? Explain whether the structure of this recurrence fits the requirements of the Master Theorem.

Q)

a) Analyze the following code snippet and express the time complexity using the Big O-notation. You must mention the time complexity for each loop.

```
int b = 0;
for (int i = n; i > 0; i -= 2) {
   for (int j = n; j > 0; j--) {
      b += i * j;
   }
}
```

```
for (int k = 1; k < n; k *= 4) {
   for (int m = 0; m < n; m += 3) {
      if (m == 10) {
         break;
      }
      b += k + m;
   }
}
return b;
```

b) You are given the following recurrence relation:

$$T(n) = 3T\left(\frac{n}{4}\right) + \Theta(3n^2)$$

Solve the recurrence relation using an appropriate method

c) Prove whether $5n^2 + 4n = \Theta(T(n))$, where T(n) is the answer you found in part Q(b).

d) Why do we ignore constants and lower-order terms when expressing the time complexity of an algorithm? Briefly explain in your own words.

Q)

Consider the following code snippet to answer questions (i) and (ii).

Function F(n):

If (n == 1):

Return True;

```
    Else:
        X = F(n/2);
        Y = F(n/2);
        Z = F(n/2);
        U = A(n);

Function A(n):
    For i := n to 1; i--:
        Return True;
```

a) Write the time complexity of the function “F” using a recurrence relation.
b) Express the recurrence relation from question (i) with asymptotic upper bound (Big-O) using any convenient method. Show your work.

Q)

Examine the following code segment and answer the following questions:

```
int funcB (int n) {
   int m = 0;
   for (int k = 1; k <= n; k += 2) {
      for (int j = n; j >= 1; j--) {
         for (int i = 1; i <= n; i += 2) {
            if (i%2 != 0) {
               break;
            } else {
               m = k - j + i;
            }
         }
      }
   }
   return m;
}
int funcA (int n) {
   if (n == 1) {
      return n;
```

```
    } else {
        x = funcA(n/2);
        y = funcA(n/4);
        z = funcA(n/6);
        w = funcB(n/2);
        return w;
    }
}
```

a) Express the time complexity of the function 'funcB' using the Big O-notation.

b) Write the recurrence relation of the function 'funcA'. (You can use your answer from question (i) if required.)

c) Solve the recurrence relation of no (ii) using any suitable method.

d) Prove (with proper explanation or a formula): $\sqrt[4]{10n^4 + 9n^3 + 8n} = \Theta(n)$

Searching Algorithms

Searching Algorithms

Q)

Consider a farm with n cows of two colors, say, white cows and black cows. The cows are arranged in their barn in such a way that all white cows are kept first, followed by all black cows. Each of the cows is assigned a serial number. You are to find the serial number of the first black cow.

a) Describe how to convert the given information into an array of integers.

b) Given the array of integers from Q(2a), present a method with a time complexity less than O(n) to solve the mentioned problem in the paragraph. Use pseudocode/ executable code/ step-by-step logical instructions to show your method.

Q)

Consider a factory where n humanoid robots work. You are to sort the robots based on some criteria.

a) Each robot can form connections with some of the other robots. That means each robot can form at most (n-1) connections. Which algorithm will be the fastest to sort the robots based on the number of connections they formed? Write the name and time complexity of the algorithm

b) Suppose, in the factory, there exists a system that can find the tallest robot in O(1) time. That means you are given a function, $find_tallest(A)$, which returns the tallest robot from a list called A, in constant time.

   Considering this information, design an algorithm to sort the robots in descending order of their heights. The time complexity of your algorithm must not exceed O(n).

   i) Present your algorithm using pseudocode/ executable code/ step-by-step logical instructions
   ii) Write the time complexity of your algorithm.

Q)

Given an array of N elements, you are required to find the peak element in it. An element is considered to be a peak if its value is greater than the value(s) of its adjacent element(s) i.e. immediate left and immediate right. If there is only one adjacent element, we take only that in consideration. And the array will contain exactly one peak element.

Now answer the following questions.

a) Determine whether each of the following arrays matches the above stem-mentioned conditions:

   i) [1, 5, 4, 3, 2]
   ii) [1, 3, 2, 5, 4]
   iii) [5, 2, 1, 3, 4]
   iv) [1, 2, 3, 4, 5]

b) Present an algorithm with a time complexity lower than O(N) to solve the stem-mentioned problem using pseudocode/ programmable code/ step-by-step logical instructions.
c) If the array contains more than one peak element, will the presented algorithm in question (ii) still be able to find any peak element? Explain briefly.

Q)

You are working with a weather monitoring station that records the temperature every hour for 12 hours during the night shift. During the night, the temperature gradually drops to reach the coldest hour, and then begins to rise again. Your task is to find the minimum temperature of the night, which corresponds to the coldest hour.

The station tracks temperature every hour over a 12-hour period. A sample pattern of the temperature reading per hour is given here:

Hour	1	2	3	4	5	6	7	8	9	10	11	12
Temperature (°C)	28	24	20	16	12	10	9	11	15	19	23	27

In this example, the temperature decreases from hour 1 to hour 7 (coldest at 9°C). Then it increases from hour 8 to hour 12.

a) Propose a linear search approach to find the minimum temperature. Present the solution with pseudocode/programmable code/step-by-step logical instructions.

b) Now propose a modification to binary search to solve the same problem. Present the solution with pseudocode/programmable code/step-by-step logical instructions.

Q)

You are given an array A consisting of N integers. Some of the numbers are positive, and some are negative. For example A : {15, -13, 3, -4, 5, -6}.

You have to organize the array in a way that all the negative numbers come to the front and all positive numbers are pushed to the back. For example, the array A from above becomes Org_A : {-6,-4,-13, 15, 3, 5}. Note that numbers don't need to be sorted here.

a) Propose a linear-time [O(N)] algorithm with constant [O(1)] space complexity. Present your algorithm as pseudocode or executable code, or step-by-step logical instructions.

b) After solving Q2(a), you want to find the index of the leftmost positive number in the organized array Org_A. Present a modification of Binary Search to achieve this. Write either pseudocode or executable code.

Sorting Algorithms

Sorting Algorithms

Q)

a) While sorting a list of 10 integers in ascending order using Quick Sort, after the first partition (using the first element as a pivot), the list looks like this : [13, 11, 19, 7, 23, 37, 29, 53, 59, 41]. Which element was the pivot before partitioning?

b) Show how the list will look like after partitioning using 13 as the pivot.

Q)

You are given an array containing N distinct integers in ascending order. You have to find out the first missing number from the increasing sequence.
For example: [3, 4, 5, 6, 9, 10]. The first missing number in this sequence is 7.
[Explanation : If A_i and A_{i+1} are two consecutive integers from the list and their difference $(A_{i+1} - A_i)$ is more than 1, then we say A_i+1 is a missing number.]

Now that you have understood the problem completely, you are trying to solve it efficiently. Your friend, Jack, said that by iterating through all the numbers from the beginning, you can find the first missing number in linear time [$O(N)$]. But, you need a more efficient one. Now, answer the questions below.

a) If N = 9 and the maximum number (obviously the last-indexed one) in the array is 15, which one among 6, 8, 10 might be the first number (obviously the minimum one) in the array?

b) Explain why you discarded the other two numbers in question no. (a).

c) Assume N = 7 and the minimum and maximum numbers are 2 and 12 respectively. You found the middle-index value to be 5. On which side (left/right) of 5, do you expect to find the first missing number? Only mention ‘left’ or ‘right’.

d) Explain why you discarded the other half of the array in question no. (c).

e) Assume N = 7 and the minimum and maximum numbers are 8 and 19 respectively. The missing elements are 11, 13, 14, 17, 18. What will be the value of the middle-index element? Determine the absolute difference between the middle-index value and the maximum number.

Q)

Renowned Progressive Rock band ‘Porcupine Tree’ released an album called 'Closure/Continuation' after about 13 years. Now as a Progressive Rock Music fan you are going to listen to the tracks of the album but in the order of their Youtube views (highest one at first, lowest one at last) instead of the order of the album tracklist.

You have chosen an Algorithm to order them as per your preference. If multiple tracks have the same views, you are going to listen to any of them the Algorithm puts first in the list after ordering. This algorithm also solves your concern about the issues regarding space that your device is facing. Porcupine Tree made the fans wait for 13 years for a new album and so you think you have the patience to wait as much time as the Algorithm may take to order the tracks.

The following table contains the list of the tracks and their Youtube views (in thousand) :

Track	Harridan	Of The New Day	Rats Return	Dignity	Herd Culling	Walk the Plank	Chimera's Wreck	Population Three
Views	15	8	11	112	33	39	88	41

a) Specify the name of the algorithm you have chosen and simulate the Algorithm to order the tracks of the album as per your preference. Show your workings in detail.

b) Suppose you want to run the algorithm again on the ordered list. Determine the run-time complexity of the algorithm in this scenario.

Q)

a) You are designing a computerized logistics system for a warehouse that handles thousands of packages daily. Each package is assigned a unique package ID and a delivery deadline (in terms of 'remaining hours' from the current time) and the packages are initially sorted according to their package_id. The warehouse management system needs to sort the packages by their delivery deadlines in ascending order to prioritize the earliest deliveries.

But there are some conditions to develop this system. The system does not have any issues related to space/memory but it needs the list to be sorted with time complexity $O(nlogn)$ where n is the number of packages to process. Moreover, if any two packages have the same delivery deadline, the package with smaller package_id comes before i.e the algorithm must be stable.

Write the name of the appropriate algorithm for this case and show the solution with pseudocode/programmable code/step-by-step logical instructions.

b) [Continues on the previous question…] After some days, the warehouse noticed a massive increase in daily orders and the system was facing memory issues to hold these massive orders. So they consulted you again. You suggested to them an algorithm which does not require any additional memory. But you also warned them that it can degrade to the time complexity of $O(n^2)$ in some cases. However, they insist the time complexity must remain O(nlogn) in all cases. So you proposed a modification to your algorithm for question (a) without increasing the time complexity

i) Name the algorithm you suggested and explain the case where the algorithm might take $O(n^2)$ time.
ii) What solution did you come up with to ensure O(nlogn)? Explain where it differs from the solution of question (a).

Q)

a) You are working on a grading system for a national standardized exam where students are marked out of 100. You need to sort millions of student scores as efficiently as possible. Speed is critical, and memory is not a major constraint. Comparisons are costly.

Which algorithm is the most suitable for the above scenario? Explain your choice.
Write the time complexity of your preferred algorithm.

b) Suppose your friend wants to implement a variant of Merge Sort called Ternary Merge Sort, where instead of dividing the array into two halves, he divides it into three equal (or nearly equal) parts and recursively sort each part, and then merge the

three sorted subarrays into a single sorted array. It is guaranteed that the input array will have a size of at least 3.

Now, he asks your help to implement the 'dividing step' of the algorithm. Present your idea with pseudocode/programmable-code/step-by-step logical instructions.

c) Does dividing into three subarrays make the algorithm asymptotically faster?

Q)

Mention the name and time complexity of the most efficient sorting algorithm among Bubble, Insertion, Selection, Merge, Quick, and Count for each of the following input scenarios, consisting of an array of N integers. For each case, you do not have any information other than the given one.

Case #1: The largest number can be in any index, but all other numbers are already sorted.
Case #2: The smallest number is 0, and the largest is 2^N.
Case #3: The array contains all the integers from 1 to N.
Case #4: The numbers are in the reverse order of how we want them to be sorted.

Divide-and-Conquer

Divide & Conquer

Q)

Your friend gave you a binary string B (meaning each character is either 0 or 1). He wanted to find out how to calculate the maximum number of consecutive 0s in that particular string.

For example,

String: 100100000111, Maximum consecutive 0s: 5
String: 1010101010101, Maximum consecutive 0s: 1

You, as an algorithm enthusiast, know that this can be solved in linear time. However, your friend asked you to propose a Divide and Conquer approach

a) Name a suitable Divide and Conquer algorithm for this task.

b) Explain how you can apply that algorithm in this scenario. Present your idea in a pseudocode/programmable code/Flowchart/step-by-step instructions.

c) Write the time complexity of your algorithm.

Q)

Inspired by Karatsuba's algorithm for multiplying integers, your friend has come up with the following divide-and-conquer algorithm for squaring n-digit numbers. Pseudocode for the algorithm is given below. Your job is to fill in the details. You may assume that n is a power of two.

Algorithm 1: KARAT-SQUARE(x)

```
   Input  : An n-digit number x
   Output: x^2
1  if n = 1 then
2  |  return x^2
3  end
4  a ← the leftmost n/2 digits of x;
5  b ← the rightmost n/2 digits of x;
6  S1 ← KARAT-SQUARE(a) ;                    /* S1 equals a^2 */
7  S2 ← KARAT-SQUARE(b) ;                    /* S2 equals b^2 */
8  S3 ← KARAT-SQUARE (_____) ;               /* fill in the blank */
9  P ← _____ ;                               /* P should equal 2ab */
10 return _____ × 10^n + P × 10^(n/2) + _____;
```

Now carefully read the algorithm above, and answer the following questions.

i) Write x in terms of a and b. Looking at lines 4 and 5 of the algorithm should help.
ii) Square your answer from (i). The formula $(m + n)^2 = m^2 + 2mn + n^2$ should help.
iii) Fill in the blank in line 8. Your answer should involve a and b in some way. The idea is to compute a square that, along with a^2 and b^2, helps you compute $2ab$. You may want to think about questions (iii) and (iv) at the same time.
iv) Fill in the blank in line 9. Your answer should involve S_1, S_2, and S_3 in such a way that P equals $2ab$.
v) Fill in the blank in line 10.
vi) Write the running time of your algorithm above.

vii) Say you have another algorithm KARAT-SQUARE-2(x) that takes an n-digit number and squares it by recursively squaring six $n/3$-digit numbers, and combining them in $O(n)$ time. Which algorithm is asymptotically faster: KARAT-SQUARE(x) or KARAT-SQUARE-2(x)?

Q)

Suppose the company, "Sparkling Ideas," has been new in the market for the past few years and you work as a financial analyst hired for this company.

You have been provided a list of the company's monthly incomes over the last year. However, the incomes are mixed with positive and negative figures, which indicates the company has faced some difficulties during certain months. The company's CFO has a special skill to identify the trend in the company's income details, which helps her make informed decisions. She believes that there is a particular set of consecutive months where the company earns the most, and she wants you to find out which one(s) it is.

Your task is to find the maximum earnings from the given monthly incomes and determine in which part of the year the company earned the most.

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
7	-15	9	-4	12	-8	3	-11	16	-2	6	-10

The figures are in millions of USD.

a) Try to think of an efficient algorithm to solve this problem and provide the result in terms of the starting and ending months, along with the corresponding total income. Show a simulation of your algorithm.

b) Calculate the time complexity of your algorithm. Show proper mathematical logic.

Q)

Imagine you're a farmer named Sam who grows different types of vegetables in several continuous fields. Each field can grow a specific type of vegetable, and based on the market value of that vegetable, each field will bring a certain profit (positive integer) or loss (negative integer). The profit or loss is estimated and noted for each field.

Sam can start farming from any field, but once he starts, he must continue farming the next fields in sequence without skipping any, until he decides to stop, because his tractor can only move to the next adjacent field and cannot skip fields.

Sam needs to choose which sequence of fields to farm to maximize his profits. The following array represents the estimated profits or losses for ten consecutive fields on Sam's farm, based on the types of vegetables that can be grown in each field and their respective market values.

[4, -18, 16, -14, 12, -1, 3, 1, -20, 15]

a) Can you help Sam decide which sequence of fields to farm for maximum profit using an efficient algorithm? What would the profit amount be? Show a simulation of your proposed algorithm. You must show which sequence of fields he needs to select in order to achieve this maximum profit.

b) Calculate the time complexity of your algorithm using proper mathematical logic. An efficient algorithm should have time complexity less than or equal to O($N * log(N)$) where N is the number of fields.

Q)

a) By showing necessary math, explain how Karatsuba's Fast Multiplication algorithm converts an N-digit multiplication to three N/2-digit multiplications.
b) Can we modify the algorithm to multiply two N-bit binary numbers? Explain how or why not

Q)

We are given an array A consisting of N integers. Some of the numbers are positive, and some are negative. For example A : {15, -13, 3, -4, 5, -6}.

We want to find the Max-Prod Subarray of the input. Just like the Max-sum Subarray, but instead of summation, we have to consider the product (multiplication) of the numbers.

We have come up with a divide-and-conquer algorithm to solve this problem. This algorithm has the same concept that you know for the Max-sum Subarray, just modified for multiplication.

```
function MaxProdSubarray(A, left, right){
    if (left == right):
        return A[left]

    mid = (left + right) / 2

    LPS = MaxProdSubarray(A, left, mid)
    RPS = MaxProdSubarray(A, mid + 1, right)
    CPS = CrossProduct(A, left, mid, right)

    return max(LPS, RPS, CPS)
}

function CrossProduct(A, left, mid, right){
    leftProd = -∞
    prod = 1
    for (i from mid to left){
        prod = prod * A[i]
        if (prod > leftProd): leftProd = prod
    }

    rightProd = -∞
    prod = 1
    for (i from mid+1 to right){
        prod = prod * A[i]
        if (prod > rightProd): rightProd = prod
    }
    return leftProd * rightProd
}
```

a) Mark the changes that you see in this algorithm than the one for Max-Sum Subarray. You can either write only those lines in your script. Or, write the whole code given above and underline the ones you want to mark. Also, you don’t have to mark the changes in variable and function names.

b) Evaluate whether the given algorithm would actually work for this problem. If your answer is yes, give a proper logical explanation. Otherwise, give an input where the algorithm will not produce the correct answer.

c) We are thinking, why not use Karatsuba's multiplication technique to make our algorithm faster? So we replace the multiplication operations (*) with Karatsuba's function. For example, $prod * A[i]$ becomes $Karatsuba(prod, A[i])$.
Write a recurrence relation [$T(N) =$ ___ $+$ ___] to express the time complexity of the algorithm after applying Karatsuba. Assume that Karatsuba's time complexity is $O(N^{1.58})$(same N as the size of the Array A).

d) Now solve your recurrence from Q(c) to find the actual time complexity. Any method is acceptable as long as you show proper work.

Graph Traversal (BFS)

BFS

Q)

You are the coach of the renowned football team “Real Madrid”. Your team is behind one goal and there is still some time left to back in the game by scoring a goal. Benzema is your main striker and if you can pass the ball to him he will give the goal for sure. Currently the ball is in your goalkeeper’s (Courtois) hand. So, your main target should be Passing the ball in minimum steps from Courtois to Benzema as the time left is very low.

Courtois can pass the ball to Rudiger or Alaba
Rudiger can pass the ball to Modric
Alaba can pass the ball to Modric, Hazard or Nacho
Modric can pass the ball to Benzema
Hazard can pass the ball to Kroos
Nacho can pass the ball to Vini
Kroos can pass the ball to Benzema
Vini can pass the ball to Benzema

Now answer the following questions.

a) Using a suitable algorithm, find the minimum number of passes your team will give to reach the ball from Courtois to Benzema. Mention the name of the algorithm, Show each step of the simulation properly to find out the minimum number of passes.

b) Find out the players who are required for this minimum number of passes scenario.

Q)

The cities in Italy are connected in such a way that their adjacency list representation looks as follows-
Rome: Milan, Venice, Florence
Milan: Rome, Florence, Bologna, Venice
Florence: Milan, Naples, Rome, Bologna, Venice
Venice: Rome, Milan, Florence
Naples: Florence, Bologna, Genoa
Turin: Genoa
Bologna: Naples, Genoa, Milan, Florence
Genoa: Turin, Bologna, Naples
Palermo:Catania, Syracuse
Catania: Palermo, Syracuse
Syracuse: Catania, Palermo

Suppose Rome suddenly gets affected with covid-19. The rest of the cities are non-affected. You are asked to find out the time steps the covid virus will take to affect all the non-affected cities if it takes one time step to travel from one city to another. So, one affected city will affect all of its neighbors in one time step.

a) Draw the graph

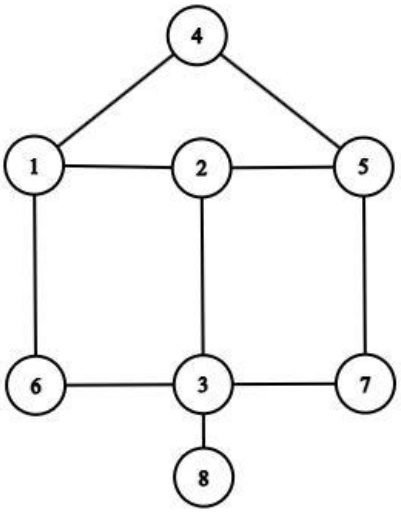
b) Mention the name of the graph traversal algorithm you think is suitable for solving the problem.

c) Simulate the suitable algorithm to find the time steps needed for each non affected city to be affected. Choose the smallest (in alphabetical order) vertex when there is a choice.

Q)

Suppose, you and your classmates are trapped inside an ancient castle during a field trip. The only way out is to solve some tasks. The tasks involve navigating the castle and resolving ancient conflicts between the spirits that guard it. Your goal is to escape through the Gate of Reason before the castle resets and traps you forever.

There are 8 rooms in the castle and you can navigate from one room to another using some corridors. The map of the castle is given below:



a) Suppose, you are in Room 4. Your first task is reaching Room 6 which is the control room and the Gate of Reason is also situated here. However, the castle is flooding and some of the corridors have been blocked. How can you determine the minimum number of steps required to reach Room 6 from Room 4 avoiding the blocked corridors?

b) Now that you have reached Room 6 and stopped the flood, all corridors are accessible once again. Only one challenge remains: unlocking the Gate of Reason to escape the castle. Each of the 8 rooms in the castle is guarded by a magical spirit. The ancient Gate of Reason only opens if the spirits can be divided into two opposing groups such that:
   - No two spirits in the same group are in conflict (the rooms they are guarding should not be connected by corridors).
   - The two groups must have the same number of spirits.

   Propose a suitable algorithm to determine whether the Gate of Reason can be opened. Present your idea with pseudocode/programmable-code/step-by-step logical instructions.

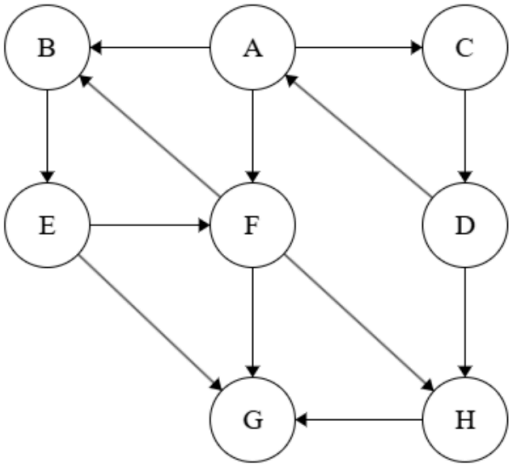
Graph Traversal (DFS) & Graph Algorithms

DFS [Spring 25]

Q)

a) For a directed graph, does the number of back edges equal the number of cycles present? Explain briefly.

b) Given a directed graph G = (V, E) represented with an adjacency list, explain how you can find whether the graph is a directed acyclic graph (DAG) or not. Your method must be as asymptotically efficient as possible.



c) Simulate Kosaraju's algorithm on the given graph Q1(b) to find the strongly connected components (SCCs). Label each SCC with the set of nodes it contains. While performing graph traversal, use node A as the starting node and break ties in alphabetical order.

d) Now, treat each SCC identified in Q1(c) as a single "super node." In this new graph, every edge from the original graph either becomes internal (and thus disappears, since it stays within an SCC) or becomes an edge connecting two different SCCs. This transformation results in a Directed Acyclic Graph (DAG). Draw this DAG.

e) Given a set of SCCs, what is the maximum number of new SCCs that can appear if we delete one edge from the DAG of these SCCs?

Consider the following algorithm

Step 1: Perform DFS on the given graph. For each node, keep track of the finishing time (the time when the algorithm finishes exploring all descendants of the node).

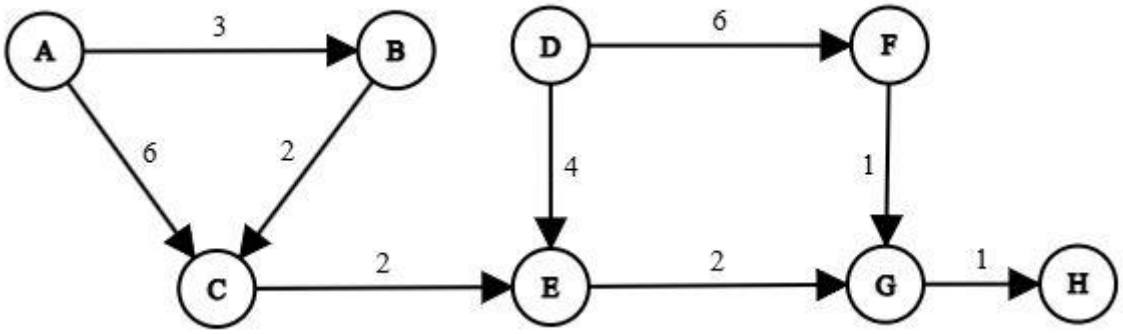
Step 2: Sort the nodes in ascending order of the finish time.

Step 3: Perform DFS in the sorted order of the nodes. For each DFS traversal, include each reachable node as one SCC.

f) Prove that the given algorithm is incorrect by providing a graph as a counterexample, on which the algorithm will fail to find the correct set of SCCs.

Q)

The country of Technovia is setting up an emergency grid that connects 8 major cities: A (Capital), B, C, D, E, F, G, H. These cities are connected by one-way roads. Each road has a traffic weight which is used to estimate the delivery delay due to congestion. Medical aid and personnel must be delivered from the capital (A) to all other cities. The map of the country is given below:



Suppose you, an algorithm enthusiast, have been assigned to determine the minimum delivery time from capital A to all other cities.

a) Which SSSP (single source shortest path) algorithm would you prefer to solve the problem?

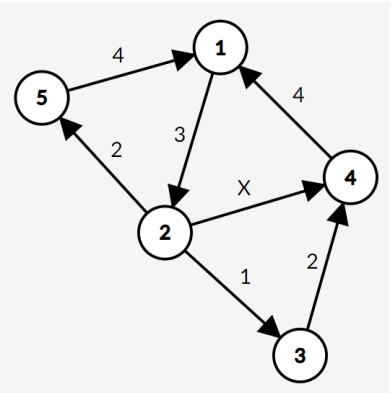
b) Simulate your mentioned algorithm with the necessary data structures to find the minimum delivery times from the capital to all other cities. Travel times are equal to the traffic weights of the roads.

Continuing from the given scenario, the Ministry of Transport enforces that while delivering medical resources, after using any high-traffic road (traffic weight > 4), the next road used in the delivery path must have a traffic weight ≤ 4. This is to prevent back-to-back gridlock.

Propose a modification to your previous algorithm (a) to determine the minimum delivery time from capital A to all other cities while strictly following the additional constraint of traffic weights. Present your idea using programmable code/pseudocode/step-by-step instructions.

[Summer 24]

Q)



a) If the edge weight X = 1, then simulate the Dijkstra Algorithm with the necessary calculation needed to find the shortest path from Source Node 1 to every node in the graph. Mention the minimum distances you could find.

b) Now, with the weight of edge (2,4) set to X = -9, prove or disprove the statement: 'This graph's single-source shortest path is unsolvable from source 1, and only Bellman-Ford can identify this issue.' Simulate/Give logical reasoning to support your answer.

c) If all roads have the same travel time, can an algorithm with time complexity O(V+E) be used to find the best route? Explain your idea in brief statements/pseudocode/flow-chart/step-by-step instructions.

[Spring 24]

Q)

Professor Bumstead needs to figure out a proper order of clothing when getting dressed. He notes down some of the dependencies which describe which garments he needs to put on before others. For example, Prof Bumstead needs to put on his shirt before he puts on his tie or his belt. He wrote down the following:

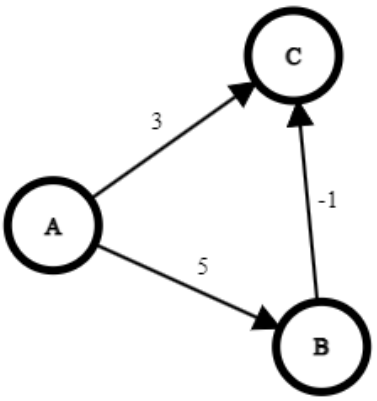
1. undershorts, watch and socks are independent
2. pants need to be put on after undershorts
3. shoes need to be put on after pants, undershorts and socks
4. belt needs to be put on after pants and shirt
5. tie needs to be put on after shirt
6. jacket needs to be put on after tie and belt

a) Write down the name of an appropriate algorithm to find the proper order of clothing items.

b) Simulate the algorithm by drawing a proper graph and determining the correct order of clothing items. If you can choose among multiple items, choose the alphabetically smallest one. For example, between "shirt" and "shoes", i comes before o, so choose "shirt".

c) Based on the clothing dependencies provided by Professor Bumstead, identify if there are any strongly connected components in the graph representation using an algorithm. Demonstrate the steps of the algorithm you are applying or justify your answer in brief.

Q)



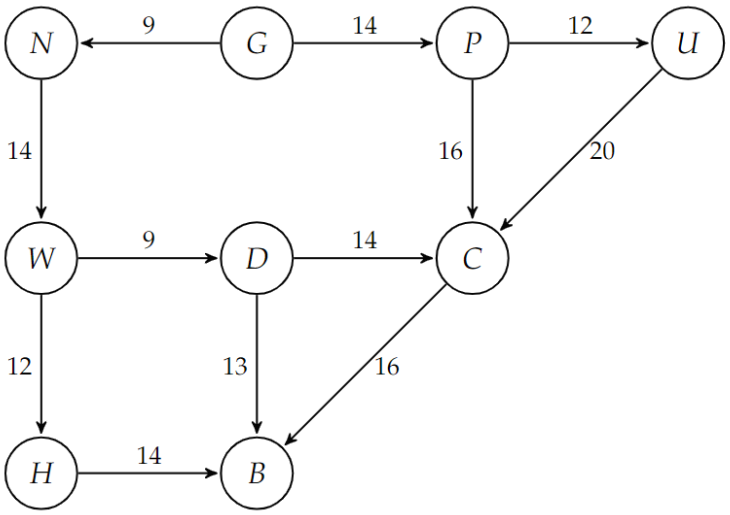
a) You have this graph where there is a negative weight on one edge. Explain why Dijkstra's shortest path algorithm will find correct answers on this graph given node A as the source. Then propose a change in the graph to make the result incorrect.

b) Someone mistakenly used Max-heap data structure instead of Min-heap to create a priority queue. So a pop/remove operation on the queue brings out the highest value element instead of the lowest one. Then this priority queue was used to implement Dijkstra's algorithm. Should it still work on any input graph? Explain your answer with proper reasoning and/or example. Assume that there will be no negative edges in the input.

[Spring 23]

Q)

Consider the network of roads connecting a set of cities given by the graph below.



a) Simulate Dijkstra's algorithm to find the weight of the shortest path from G to B.

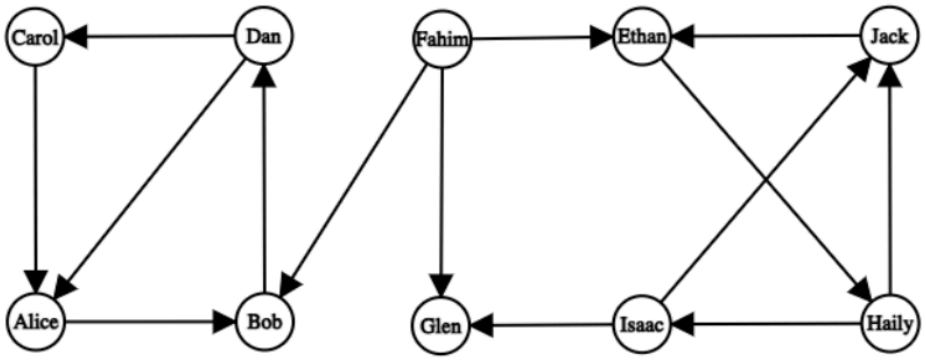
b) There is a proposal to add one new road to this network, and there is a list of five pairs of cities between which the new road can be built. Each such potential road has an associated length. As a designer for the public works department, you are asked to determine the road whose addition to the existing network would result in the maximum decrease in the distance from G to B. One way to do this would be to add

each of the five edges separately, run Dijkstra's algorithm five different times, and then compare the results. But can you do this by invoking Dijkstra's algorithm only two times (and possibly doing a linear amount of extra work)?
Present your idea with a pseudo-code/flowchart/step by step instruction/algorithm explanation.

[Summer 23]

Q)

Among a group of 10 friends, it's not feasible for each individual to visit the houses of all the others. With an upcoming tournament in mind, the friends need to establish groups. The criterion for forming these groups is that a friend must be capable of visiting the houses of every other member in their group. The task involves calculating the count of viable groups following this criterion. A group has to consist of a minimum of two people.



a) The end result will showcase the list of groups and the members. Name the algorithm tailored for this situation.

b) Show a simulation of your chosen algorithm using the graph above. Write the number of groups that can be formed and the members of each group.

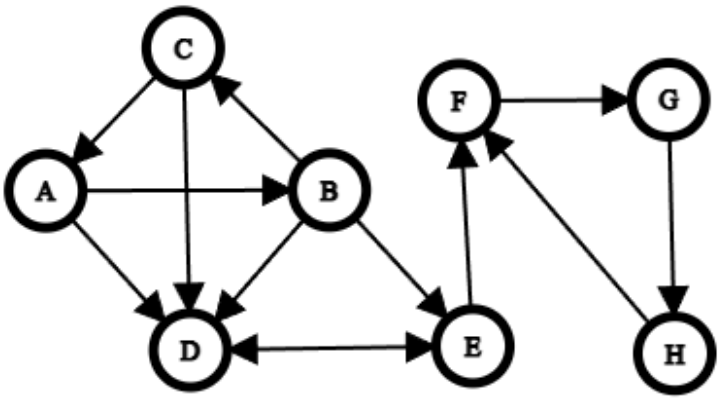
c) Those who are incapable of forming a group will be identified as individuals ineligible for tournament participation. Identify those people if there are any.

d) We also want to find the person who can visit the most houses. Propose an algorithm to implement this. Present your algorithm with a pseudocode/flowchart/step-by-step instructions.

[Fall 25]

Q)

Consider the directed graph given below.



a) Simulate a DFS on the graph to find the Discovery and Finish times for each vertex. Start with Vertex ‘A’.

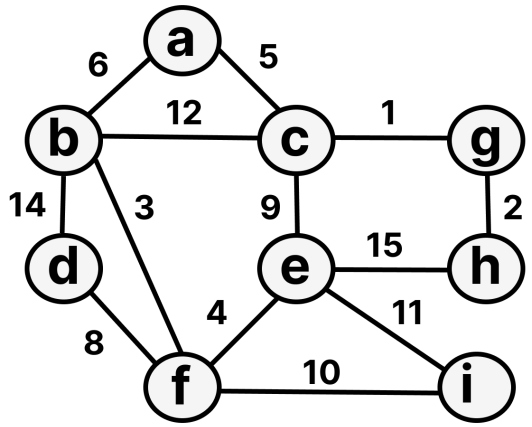
b) Now, simulate an appropriate algorithm to identify the Strongly Connected Components. You can reuse your answer to Question (a) to save time.

c) Find out the minimum number of edges to remove from the graph, so that there exists a topological order among the vertices. Explain your reasoning

d) If we consider each SCC [that you found in Question (b)] as a single vertex, will there be any remaining cycles in the graph? Draw the graph to justify your answer.

Q)

A country has several cities that need to be connected by highways. Each edge in the graph represents a potential highway connection between two cities, with weights indicating the construction cost of building that highway.



a) Assume all highways have been constructed as shown in the graph. The government has installed toll plazas on each highway. The edge weights now represent the toll fees for using each highway. You're planning a trip from a source city to a destination

city and want to minimize your total toll cost. You have a special coupon that allows you to skip paying the toll on only one highway (you can use it at most once during your journey).

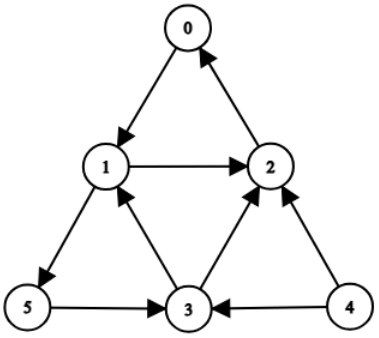
Your friend proposed a modified approach: First, apply standard Dijkstra's algorithm to find the shortest path from source to destination while keeping track of which edge has the highest toll cost on this path. Then, use the coupon on that edge to skip its toll cost. According to your friend, this will guarantee the minimum total toll cost from source to destination.

Do you agree that this approach will guarantee the shortest path in all cases, regardless of how the cities are connected, given that all toll fees are positive? If yes, provide proper reasoning. If not, provide a counterexample showing where this approach will fail.

[Fall 24]

Q)

You have just joined a new university and are exploring the campus, which consists of several buildings connected by one-way bridges. These bridges form a directed graph, where each node represents a building and each edge represents a bridge between two buildings. You aim to analyze how the buildings and bridges are connected to understand the campus layout better and improve your navigation between classes and facilities.



a) Mention which graph representation will let you check whether any two buildings have a bridge between them in O(1) time.

b) From Figure 1

- Simulate an algorithm to find the Strongly Connected Components (SCC).
- Suppose a bridge gets blocked. Explain how you can efficiently calculate the updated SCCs from b(i) without rerunning the algorithm on the entire graph.

c) A building is called "*Accessible*" if it's possible to start from there, visit some (or all) of the other buildings, and eventually return to that building.

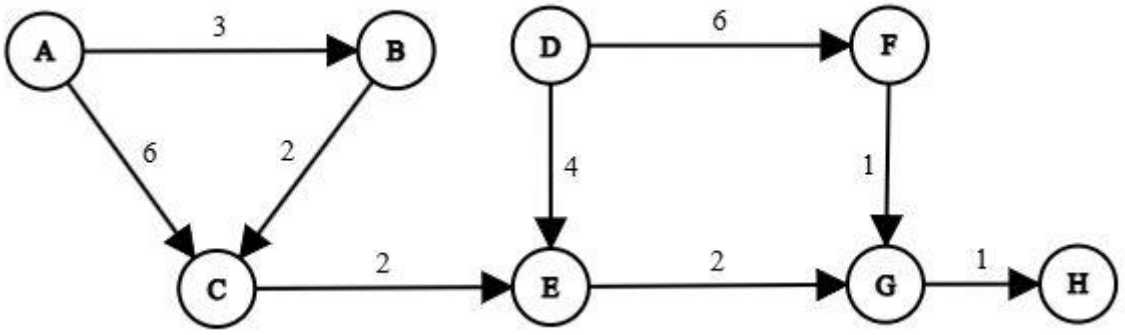
Propose an algorithm with a pseudocode/programmable code/step-by-step logical instructions to determine whether every building on the campus is *Accessible*. The algorithm should run in at most $O(V \times (V + E))$ time.

Greedy Algorithms

Greedy [Spring 25]

Q)

The country of Technovia is setting up an emergency grid that connects 8 major cities: A (Capital), B, C, D, E, F, G, H. These cities are connected by one-way roads. Each road has a traffic weight which is used to estimate the delivery delay due to congestion. Medical aid and personnel must be delivered from the capital (A) to all other cities. The map of the country is given below:



a) The Ministry wants to design a minimum-traffic infrastructure plan that connects all the cities (an MST of the graph). Will Kruskal's or Prim's MST finding algorithm work properly for the given one-way roads? If yes, justify your answer. Otherwise, propose necessary modifications to the graph so that the algorithm works.

b) While finding the MST of the graph, we want to enforce a rule that 'No city can have more than two (02) roads'.

c) Propose a modification to the MST algorithm (either Kruskal's or Prim's) to accommodate this requirement. Present your idea using programmable code/pseudocode/step-by-step instructions

Q)

The final exam has just ended, and you are on a well-deserved trip to a remote area. During your journey, you have collected different types of digital content that you want to share with your friends and family. Each Content Type has a Total Duration and a Value. Total Duration measures the content's pre-compression size, while Value reflects its importance or meaning to the audience.

Due to bandwidth constraints, one of your friends suggested using Huffman encoding, a lossless compression technique that assigns shorter binary codes to more voluminous (i.e., longer duration) items.

Content Type	Total Duration (Seconds)	Value
A	13	20
B	11	14
C	9	6
D	8	17
E	6	11
F	5	7

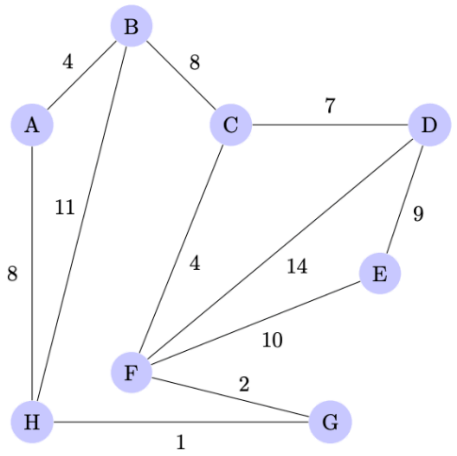
G	4	8

a) Construct a Huffman Tree based on the given Total Duration to assign optimal binary codes to each content type.

b) Calculate the Total Number of Bits (duration * code length) required to transmit the full content of each type.

[Summer 24]

Q)



a) For the given transportation network, find a configuration connecting all regions with minimal cost. Demonstrate the steps and the algorithm you used to select connections.

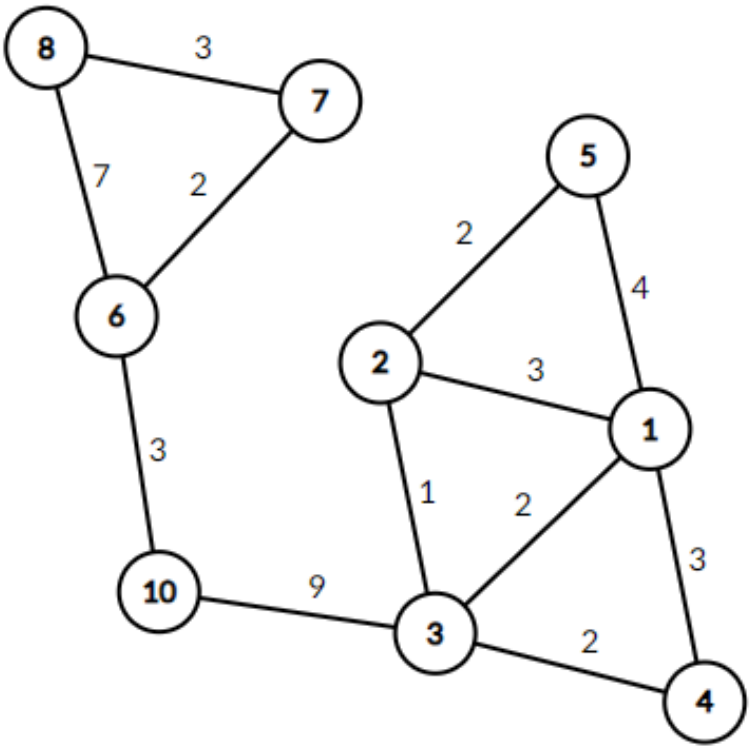
b) Based on the result from part (a), calculate the total cost of your configuration and explain in short, why it represents the minimum cost required to connect all regions.

c) Now, imagine the road between Region C and Region F became unavailable due to maintenance. How would this impact overall connectivity? Propose an alternative solution to maintain network connectivity with minimal disruption, and explain your reasoning.

[Spring 24]

Q)

Construct the minimum spanning tree from the graph below. Present your work with proper details/data structures. Finally, draw the MST and write the total cost of that tree.



Q)

Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit has been set to 9. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses. Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	9	7	15	12	16

a) Wye wants to send a message "Successful". Instead of fixed-length coding, he is willing to send the message using Huffman Coding. Show simulation of Huffman coding to encode his message.

[Spring 23]

Q)

Following are the codes generated from a text for a Huffman tree construction.

H - 1000 *o* - 1001 <*space*> - 1010 *S* - 1011	*u* - 000 *d* - 001 *n* - 010	*e* - 011 *l* - 110 *t* - 111

You are also given the following information:

- The frequency of each leaf node except *e*, *l*, and *t* is 1.
- The left and right child nodes of the root have frequencies 5 and 8 respectively.

Now answer the following questions.

a) Suppose in a Huffman tree, the distances from the root to the pair of leaves denoting the letters *k* and *b* are 5 and 2 respectively, which letter between them is more frequent in the original text? Just mention the letter.

b) Draw the Huffman tree from the given coding table above.

c) Continuing on Q(b), what are the frequencies of *l*, and *t* in the original text? Just mention the frequencies.

Your lazy friend, Tom, used a very simple encoding method for compressing a text file. The text file contains the string : *Sudden Hello*. He used a constant number of bits for encoding each of the *n* distinct characters in this text. The constant is $ceil(log_2 n)$ in this case. On the other hand, you have compressed the text file using your own huffman tree from Q(b).

d) Mention how many bits each of the characters in Tom's encoding scheme contain.

e) Compare between the number of bits needed to decode the above-mentioned string using your friend's scheme and your Huffman tree from Q(b).

[Summer 23]

Q)

The white council once decided to install bidirectional rail tracks to connect several realms of middle earth. Upon their request, the elves and the dwarves devised a plan. The following table lists the tracks that can be constructed.

From	To	Length of the track	Construction cost	Construction time
Arnor	Fangorn	25	70	80
Arnor	Lindon	20	40	30
Arnor	Mirkwood	30	80	70

Fangorn	Gondor	10	30	40
Fangorn	Lindon	40	90	110
Fangorn	Mirkwood	20	60	50
Gondor	Harad	25	100	90
Gondor	Lindon	45	110	100
Gondor	Mirkwood	30	50	60
Gondor	Rhun	40	120	130
Harad	Mirkwood	45	150	180
Mirkwood	Rhun	25	180	150

As there is only one engine available for the task, simultaneous construction of multiple tracks is not possible. Going through the plan, the council decided that the following tracks must be constructed.

- Gondor to Lindon
- Arnor to Mirkwood

However, due to a financial crisis, the council wants to connect all the realms with minimum total construction cost. Now answer the following questions.

a) Which other tracks should they select for construction? Show a simulation of your procedure.

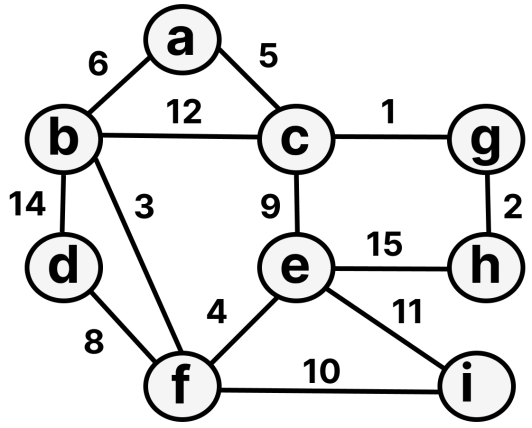
b) Write the total construction cost of the selected tracks.

c) Calculate the total construction time required if no two tracks are constructed simultaneously.

[Fall 25]

Q)

A country has several cities that need to be connected by highways. Each edge in the graph represents a potential highway connection between two cities, with weights indicating the construction cost of building that highway.



a) The government wants to ensure all cities remain connected while minimizing the total construction cost. Simulate a suitable algorithm to determine which highways should be built. Report the total minimum construction cost.

b) Is it possible for a road network (not necessarily the one given above) to have multiple different configurations of highways, each resulting in the same minimum total construction cost? If yes, provide a simple example graph where this occurs. If not, justify your answer with proper reasoning.

Q)

You are designing a lossless data compression module for a real-time monitoring system. After analyzing system logs, the following character frequency distribution is obtained:

Character	Frequency
D	5
A	30
R	12
B	18
C	10
E	4
U	8

a) Even though all character frequencies are unique, one can obtain different Huffman Trees for above characters. Is this possible? Explain whether the bit patterns and the optimality of the code will be affected by this.

b) Simulate the Huffman Encoding algorithm for the given characters. Construct the huffman tree, generate Huffman code and bits required for each character. (consider left edge as 0, right as 1).

c) If you were to send the message “BRACU” with an 8-bit encoding scheme, how many extra bits would have been required compared to Huffman encoding? Ignore the size of the code table.

d) “In Huffman Encoding, a higher-frequency symbol always gets a shorter codeword than a lower-frequency symbol.” Is this statement always true? Explain.

[Fall 24]

Q)

Consider the following scenario

You are designing a data compression system for a critical database server. You have analyzed the following character frequencies from the database logs:

Character	Frequency
A	45
B	13
C	12
D	16
E	9
F	9
G	12

a) Construct a Huffman tree for the given characters and their frequencies. Additionally, calculate the space saved compared to using fixed 3-bit encoding for each character. Show your work step by step

b) Write the Huffman Code for each character.

c) Explain how greedy property is used in Huffman coding and how it provides an optimal code.

d) Two friends are tasked with building the given Huffman tree. While both friends use priority queues in the process,

- friend 1 inserts the newly created nodes at the end of the same frequency nodes.
- friend 2 inserts the newly created node at a random position between the same frequency nodes.

Explain which technique will provide more optimal code and why. If they are bound to provide equally optimal code, explain why

Dynamic Programming

DP [Spring 25]

Q)

a) String1: <Your ID>
String2: _____

i. Create String2 in a way that the following constraints match. Write the string.
- Length of String2 = 8 (eight)
- Length of LCS(String1,String2) = 5 (five)
- There are exactly three distinct LCS (of length = 5)

ii. Write all three LCS strings (just write, no need to show any work).

b) String1: <Your ID>
String2: “20251905”

iii. Find the length of the LCS of these two strings. You can use a memory table or a recursion tree. Show your work.

iv. Using your work from the previous question, find the LCS string.

Q)

The final exam has just ended, and you are on a well-deserved trip to a remote area. During your journey, you have collected different types of digital content that you want to share with your friends and family. Each Content Type has a Total Duration and a Value. Total Duration measures the content's pre-compression size, while Value reflects its importance or meaning to the audience.

Due to bandwidth constraints, one of your friends suggested using Huffman encoding, a lossless compression technique that assigns shorter binary codes to more voluminous (i.e., longer duration) items.

Content Type	Total Duration (Seconds)	Value
A	13	20
B	11	14
C	9	6
D	8	17
E	6	11
F	5	7
G	4	8

a) Due to bandwidth constraints, you are now trying to select which content to send and how much of each. You can shorten any content, but its value drops proportionally if you do so. Your goal is to choose the types and lengths of the contents that give you the highest total value without going over 30 seconds. Determine which content to send and how long each part should be.

[Summer 24]

Q)

You are given two strings, 'cbabdcb' and 'badcbc', and tasked with finding a common sequence with the longest length. Initially, you considered a recursive solution, but person 'X' suggested using a dynamic programming approach instead.

a) Find the longest common sequence using the approach proposed by person ‘X’. Show the necessary simulation to find the answer.

b) Additionally, you want to send these strings as an encoded message. After merging them into one string, 'cbabdcbbadcbc', now you need an efficient text-encoding algorithm with minimal space requirement.

- Simulate the encoding process,
- construct any needed tables or trees,
- generate the codeword for each character,
- show the encoded message and
- the total number of bits required to store it.

Q)

You are now attending the CSE221 final exam and there is a limited time to solve the questions. Each question requires a different amount of time and offers varying marks. After answering some questions, you suddenly realized that only 6 minutes remained. You aim to maximize the total marks you can score, given the constraints on time.

Questions and their properties:

- Question 4: Time = 4 minutes, Marks = 7
- Question 5: Time = 2 minutes, Marks = 9
- Question 6: Time = 3 minutes, Marks = 10
- Question 7: Time = 5 minutes, Marks = 8
- Question 8: Time = 6 minutes, Marks = 12
- Question 9: Time = 1 minute, Marks = 3

a) If your faculty doesn’t give partial marks for incomplete answers, which combination of questions maximizes your total marks without exceeding the time limit? Hurry up, you need a quick, efficient approach—no time for brute force. Show a full detailed calculation.

b) Fortunately, your faculty enters the hall and announces a new rule: partial marks will be given for partially completed answers. If you can solve part of a question, you’ll get marks proportional to the time you spend on it. For example, solving half of a 4-minute question gives you half the marks.

Now, with this new opportunity, how does your strategy shift? How will you maximize your score within the time limit, factoring in that you can now earn something even if you don’t fully complete a question? Calculate the maximum marks you can score with this fortunate announcement. Did you score more than the previous scenario?

c) How does Dynamic Programming have a lower time complexity than brute force or recursive solutions? Explain briefly.

[Spring 24]

Q)

Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit has been set to 9. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses. Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	9	7	15	12	16

a) Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. Show your work in detail, then write down the selected courses.

b) Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. Determine whether Zed could select more valued courses than Wye.
Note that, selecting a course still means taking the whole, with full credits.

[Spring 23]

Q)

Two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the museum are too precious, and taking a fraction of any item won't sell on the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Royal Treasury with an empty knapsack weighing a total of 7 kg each. Even though both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Royal Treasury Museum are listed below.

Objects	Diamond	Jewelry	Sculpture	Painting	Gold Crest
Profit ($)	3	4	12	9	12
Weigh (kg)	1	2	8	4	5

a) Simulate your dynamic programming algorithm to find the maximum profit Denver can make.

b) From your memory/dp table, find out which items would he take to make this amount of profit?

c) Does Nairobi's belief remain valid after the robbery? Prove it.

[Fall 25]

Q)

This is the LCS Table for two strings $X=X_1X_2X_3X_4$ and $Y=Y_1Y_2Y_3Y_4$

i\j	-	X_1	X_2	X_3	X_4
-	0	0	0	0	0
Y_1	0	0	0	0	0
Y_2	0	0	0	**1**	1
Y_3	0	0	1	1	2
Y_4	0	1	1	**1**	2

a) What is the direction of cells where values are underlined, i.e which adjacent cells they came from?

b) If String Y = <Rightmost four digits of Your Student ID>, construct a correct String X.

c) The formula to find LCS can be implemented both iteratively and recursively. Which one would be more efficient? Explain your answer briefly.

[Fall 24]

Q)

a) Can merge sort be written as a dynamic programming algorithm? Explain your answer

b) We are calculating the longest common subsequence (LCS) between two strings, S= X1X2X3X4 and T = Y1Y2Y3Y4. To do this, we fill an array C where Ci,j represents the length of the LCS between the prefix of length i from S and the prefix of length j from T.

Some of the entries in the array C are currently masked using different symbols other than integers as shown below.

	Y_0	Y_1	Y_2	Y_3	Y_4
X_0	0	0	0	0	0
X_1	0	0	0	0	1
X_2	0	0	0	1	1
X_3	0	0	1	1	♡
X_4	0	1	♣	♠	♢

i. What does the cell $C_{0,0}$ represent?

ii. What can be said about the characters X_1 and Y_4?

iii. What can be said about the characters X_2 and Y_4?

iv. What is the value of ♡?

v. What is the value of ♣?

vi. When we are filling up the i-th row of our dynamic programming table C, what rows do we need to have access to?

vii. Given the observation above can we optimize our space complexity further?